MGFEI-Net: Multiscale Grouping Feedback Embedded Integrated Network for Panchromatic, Multispectral, and Hyperspectral Image Fusion

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***Abstract*— The spaceborne hyperspectral (HS) imagery with fine spectral information has broad application aspects; however, the low spatial resolution (LR) has limited the potential application values. Over the past few decades, a general strategy to improve the spatial resolution of the HS is to fuse the LR HS with an auxiliary moderate spatial resolution (MR) multispectral (MS) or a high spatial resolution (HR) panchromatic (PAN) image. However, most of the existing methods mainly focus on two-sensor fusion with the LR HS and MR MS images (i.e., MS–HS fusion) or the LR HS and HR PAN images (i.e., PAN–HS fusion). How to comprehensively combine the complementary spatial and spectral advantages of the LR HS, MR MS, and HR PAN observations, to obtain the optimal high-fidelity HR HS image is interesting and challenging. In this article, we propose a multiscale grouping feedback embedded integrated fusion network (MGFEI-Net) for the LR HS, MR MS, and HR PAN images. Specifically, an attention-based hybrid-scale integrated module (HIM) is designed by considering the spatial scale diversity of the HR PAN, MR MS, and LR HS images. Moreover, a multiscale grouping feedback embedded (MGFE) module with a top-to-bottom manner is proposed to capture more useful spatial–spectral features. Experiments were performed on the simulated and real datasets. Moreover, the robustness of the proposed PAN–MS–HS fusion under different large spatial resolution ratios (such as 8, 16, 32, and 64) was analyzed. The experimental results demonstrated the competitive performance of the proposed method.**

***Index Terms*— Deep learning (DL), hyperspectral (HS) image, integrated fusion, multispectral (MS) image, panchromatic (PAN) image.**

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I. Introduction

DUE to physical limitations of the signal-to-noise ratio, data storage and transmission, and other factors, existing remote sensing satellite images have to make a fundamental tradeoff between the spatial and spectral resolutions. For example, hyperspectral (HS) imagery has a high spectral resolution with hundreds of bands; however, it generally features a low spatial resolution (LR). The multispectral (MS) imagery is generally characterized by a few wider spectral bands, such as four bands for IKONOS MS and eight bands for WorldView-2 MS. Fortunately, the spatial resolution of MS is generally higher than that of HS. The panchromatic (PAN) imagery has the highest spatial resolution compared with MS and HS; however, it only has a single band. Therefore, how to comprehensively combine the complementary spatial and spectral advantages of the high spatial resolution (HR) PAN, the moderate spatial resolution (MR) MS, and the LR HS images, to obtain the optimal high-fidelity HR HS fused image at the highest spatial resolution of the HR PAN is a meaningful but challenging problem.

To date, various fusion methods have been proposed to improve the spatial resolution of the LR HS by the auxiliary PAN or MS images. Existing methods can be divided into three major categories: PAN–HS fusion, MS–HS fusion, and PAN–MS–HS fusion.

The PAN–HS fusion aims to improve the spatial resolution of the LR HS by the HR PAN to obtain the HR HS. Earlier PAN–HS fusion methods were directly borrowed from the PAN–MS fusion (i.e., the typical pansharpening) [1], [2], [3], due to their similar processing. Correspondingly, the PAN–HS fusion is also known as HS pansharpening. The typical PAN–HS fusion methods include component substitution (CS)-, multiresolution analysis (MRA)-, CS-MRA hybrid-, Bayesian-, and deep learning (DL)-based methods [4], [5], [6]. The CS-based methods rely upon the projection of the

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spectral bands of HS into another space, based on intensity-hue-saturation (IHS) [7], [8], principal component analysis (PCA) [9], and so on, to separate the "spatial" and "spectral" components, and then, the LR "spatial" component of the HS is substituted by the HR PAN to obtain the HR HS image. The MRA-based methods are based on the injection of the high spatial structures, extracted from the HR PAN by wavelet [10] or Laplacian filter [11], etc., into the resampled HS image, to obtain the HR HS image. In the former two categories, the CS-based methods generally generate fused results with better spatial structures but spectral distortion. In contrast, the MRA-based approaches perform better in spectral preservation but are more sensitive to registration errors and may cause spatial distortions. In this case, the CS-MRA hybrid methods were proposed to achieve a balance between spatial sharpness and spectral fidelity, such as the typical guided filter PCA (GFPCA) method [12]. In addition, the Bayesian-based methods [13], [14], [15] with rigorous mathematical logic were developed, and they generally show a higher performance in spectral fidelity. However, it is generally time-consuming due to the iterative optimization of the fusion energy function. In recent years, DL with strong automatic advanced feature extraction and nonlinear representation ability has been attracting ever-increasing attention in PAN–HS fusion. Based on the pansharpening neural network (PNN) [16] on simple three convolutional layers, He et al. [5] proposed the HyperPNN by introducing a spectral prediction structure to solve the inconsistent spectral range problem between PAN and HS images. Li et al. [17] proposed detail-based deep Laplacian pansharpening (DDLPS) to improve the spatial resolution of HS imagery. With the successful application of the attention mechanism in various visual tasks [18], [19], [20], [21], Zheng et al. [6] proposed an HS pansharpening method based on a dual attention network by jointly exploring the spatial and spectral dependence of HS images, to make the network pay more attention to useful features and improve learning ability.

The MS–HS fusion aims at improving the spatial resolution of the HS by corresponding MS. The PAN–HS fusion can be regarded as a particular case of the MS–HS fusion, for which the HS bands are segmented into several subsets, and each spectral subset can be fused with the corresponding MS band by the PAN–HS fusion algorithm. Most MS–HS fusion methods can be roughly categorized as follows: CS-, MRA-, matrix factorization (MF)-, constrained optimization model (COM)-, tensor-, and DL-based methods. The implementation of the CS- and MRA-based methods is similar to the PAN–HS fusion by regarding the MS as consisting of multiple single-band PAN images. The MF-based methods [22], [23] have been actively investigated to solve the MS–HS fusion problem. It is assumed that the mixed pixels can be decomposed into spectral endmember and abundance. For the MF-based fusion methods, the endmember is obtained from the HS image and the abundance stems from the MS image. For example, Kawakami et al. [24] proposed an MF-based MS–HS fusion method in two stages: the spectral basis was first obtained by unmixing the HS image, and then it was in conjunction with the MS (RGB) to produce the fused image. In addition, the state-of-the-art coupled nonnegative matrix factorization (CNMF) method proposed by Yokoya et al. [23] was popular and showed competitive performance. The COM-based methods are based on the optimal solution of a variational energy functional, constructed on the observation model and prior regularization. These methods include fast fusion based on the Sylvester equation (FUSE) [15] and maximum a posterior-stochastic mixing model (MAP-SMM) [25]. The tensor-based methods solve the MS–HS fusion problem from the viewpoint of the tensor. It regards the HS image as a tensor by a more thorough consideration of the spatial and spectral correlations. For example, Li et al. [26] proposed that the input HS images were approximated by a tensor with four factors, including a core tensor multiplied by dictionaries of three modes. DL-based MS–HS fusion methods are attracting increasing attention. Zheng et al. [6] introduced a deep network to implicitly learn a general image prior and then proposed a two-stage super-resolution network to recover the fused image from coarse to fine in an unsupervised manner. The divide-and-conquer framework (BDCF) proposed by Sun et al. [27] is dedicated to achieving high-fidelity fusion in the full spectrum of HS. Zheng et al. [28] proposed an unsupervised HyCoNet fusion network composed of three autoencoder structures to solve the MS–HS fusion problem in the case of unknown point spread function (PSF) and spectral response function (SRF).

The above fusion methods focus on the spatial resolution enhancement of the LR HS by an auxiliary PAN image (i.e., the PAN–HS fusion) or an MS image (i.e., the MS–HS fusion). The PAN–MS–HS fusion is dedicated to comprehensively integrating the complementary spatial and spectral advantages of the HR PAN, MR MS, and LR HS images, to obtain the HR HS image. There are two fusion strategies: 1) step-by-step fusion of the LR HS, MR MS, and HR PAN images and 2) the integrated PAN–MS–HS fusion. The step-by-step fusion is the commonly used method [29], [30]. That is, the LR HS image is gradually fused step by step with the MR MS and HR PAN images. For example, the LR HS image can be first fused with the MR MS image by MS–HS fusion methods to obtain the MR HS image, and then the output MR HS image is fused with the HR PAN based on the PAN–HS fused algorithms to obtain the final HR HS image. However, on the one hand, how to select the optimal MS–HS fusion and PAN–HS fusion methods at different stages, especially for different scenes, is controversial and time-consuming; on the other hand, step-by-step fusion may introduce error propagation and accumulation. The integrated PAN–MS–HS fusion is dedicated to collaboratively combining the PAN, MS, and HS images, to make full use of their complementary spatial and spectral advantages. Yokoya et al. [31] proposed the coupling matrix decomposition (CNMF) method for PAN, MS, and HS images. First, the HS and MS images were unfolded as a basis matrix (endmember) and a coefficient matrix (abundance), respectively. Then, the HS endmembers and MS abundance matrix were used to estimate the HR abundance map from PAN. Finally, multiply the HS endmembers with the HR abundance matrix to obtain the HR HS. Shen [32] proposed a maximum a posteriori

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(MAP)-based framework to solve the fuse problem of more sensors, but the method was only validated using simulated images. Meng et al. [33] further proposed a modified MAP-based PAN–MS–HS fusion method, and the experiments on real ETM+ PAN, MS, and MODIS images were implemented. In addition, Tian et al. [34] proposed the COM-based HyperFusion to simultaneously fuse PAN, MS, and HS images.

Overall, compared to the PAN–HS fusion and MS–HS fusion, the PAN–MS–HS fusion can make full use of more complementary spatial and spectral advantages of the HR PAN, MR MS, and LR HS, to obtain the HR HS at the highest spatial resolution of HR PAN image. However, how to effectively integrate their complementary information to obtain the high-fidelity HR HS fused image is challenging. On the one hand, compared with the two-sensor fusion of PAN and HS or MS and HS images, the multisensor PAN, MS, and HS observations provide more complementary spatial and spectral advantages; however, on the other hand, they are featured with a more complex and highly nonlinear relationship. Most existing PAN–MS–HS fusion methods are mostly founded on the linear model [23], [29], [35], which usually causes spectral distortion, making it difficult to achieve high-precision fusion effects. Deep convolutional neural networks (CNNs) achieve state-of-the-art performance in various computer vision, especially in remote sensing image fusion, which has become a new research trend [36], [37], [38], [39], [40]. Correspondingly, a handful of DL-based PAN–MS–HS integrated fusion methods were investigated. To the best of our knowledge, Li et al. [41] was the first study to investigate how to adopt DL in the effective fusion of the PAN, MS, and HS images. Given the realistic limitations of imaging devices, Meng et al. [42] proposed an interesting integrated fusion for PAN, MS, and HS remote sensing images with different swath widths. In addition, it should be noted that the PAN–MS–HS fusion provides a potential for a significant improvement of the HS spatial resolution; however, most of the existing fusion methods are focused on two or four times in the spatial resolution ratio, and this cannot take maximum advantage of the multisensor PAN, MS, and HS images. Therefore, how to achieve the PAN–MS–HS fusion with more challenging large spatial resolution ratio (such as 8, 16, 32, and 64 times), to substantially improve the spatial resolution of HS with high fidelity is an interesting but potentially rewarding problem.

In this article, we propose an integrated PAN–MS–HS fusion network, namely, multiscale grouping feedback embedded integrated fusion network (MGFEI-Net). The main contributions of this article are as follows.

1) We propose an integrated fusion of the HR PAN, MR MS, and LR HS images by a joint end-to-end spatial–spectral learning network, to generate the high-fidelity HR HS images. As far as we know, this is one of the few efforts to solve the integrated PAN–MS–HS fusion problem in DL theory.
2) In the proposed method, a hybrid-scale integrated module (HIM) is designed by considering the spatial resolution difference of the PAN, MS, and HS images. In addition, to capture more spatial and spectral details at different scales without increasing the memory burden, a multiscale grouping feedback embedded (MGFE) module with an interesting top-to-bottom feedback connection is proposed.
3) The experimental results indicated that our proposed network has a more effective performance for enhancing the HS spatial details while preserving the spectral information than other compared methods. Ablation studies were also provided to improve model's comprehensibility concerning typical challenges in ubiquitous HS quality improvements. Especially, we demonstrate the PAN–MS–HS fusion method can learn more useful features than PAN–HS by introducing the MS image. In addition, the proposed method is discussed and analyzed with different spatial resolution ratios, such as 8, 16, 32, and 64, to explore the robustness of the fusion model.

The structure of this article is organized as follows. The proposed fusion method is described in Section II. The experimental results and analysis are carried out in Section III. The discussion is presented in Section IV. The conclusion is given in Section V.

II. Proposed Method

A. Motivation

In this article, an integrated fusion method for PAN, MS, and HS images is proposed. The proposed method aims to make full use of the complementary spatial and spectral advantages of the multisensor HR PAN, MR MS, and LR HS images, to obtain the desired HR HS image with the highest spatial resolution of the HR PAN and the high-fidelity spectral information of the LR HS image. To achieve this aim, two challenging problems should be solved.

1) Challenge 1: Compared to two-sensor fusion, such as the PAN–MS fusion, MS–HS fusion, and PAN–HS fusion, the multisensor PAN, MS, and HS observations are characterized by more complex and nonlinear relationship; how to construct an effective fusion framework to achieve the collaborative and integrated fusion, to avoid the error accumulation and spectral distortion of step-by-step fusion.

2) Challenge 2: The PAN–MS–HS fusion provides a potentiality for the spatial resolution improvement of the LR HS image under a large spatial resolution ratio, and this is an interesting but potentially rewarding problem; however, how to design a high-fidelity fusion model by comprehensively integrating the complementary spatial and spectral advantages of the PAN, MS, and HS images with large spatial resolution ratio, to obtain the high-fidelity HR HS image is challenging.

In this article, we propose an integrated PAN–MS–HS image fusion method, namely MGFEI-Net. To solve Challenge 1, we proposed an HIM, to extract the scale-invariant object at a different resolution and adopt an attention-gating mechanism to screen and aggregate specific features from the input. Thus, our model can effectively mine the spatial and spectral relation among the input HR PAN, MR MS, and LR HS images, and produce accurate feature fusion. For Challenge 2, we investigate the MGFE module to promote the network to learn stronger feature representation for multiscale

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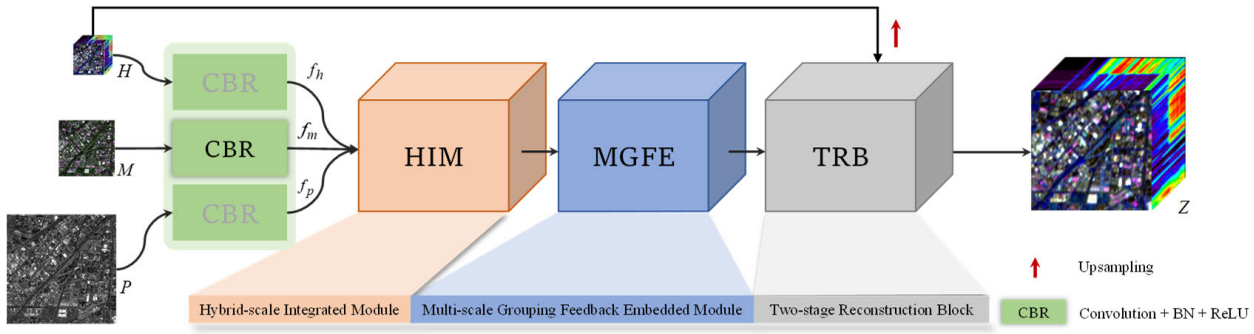


Fig. 1. Overview of the proposed MGFEI-Net. The HIM is designed to construct a unified feature space among PAN, MS, and HS images. The MGFE module is to enhance multiscale feature representation and fuse the crosslayer features. Finally, TRB is to refine the final HS feature.

spatial information without increasing memory burden while maintaining competitive spectral performance. In the module, the hierarchical feature split operator, grouped multiscale dilated block, residual channel-wise attention (RCA) block, and feedback embedded structure are designed. Then, two "Conv-BN-ReLU" (CBR) layers and 3×3 kernels are used to directly reconstruct the integrated spatial and spectral features from the output of grouping feedback embedded network. Finally, for spectral information preservation, the upsampled HS was added to the deep network. Through the proposed method, the high-fidelity HR HS image with the highest spatial resolution of the HR PAN can be obtained.

B. Proposed Fusion Model

Let the desired HR HS image be denoted as $\mathbf{Z} \in \mathbb{R}^{H \times W \times C}$, where H and $W$are the height and width, respectively, and C is the number of spectral bands. Let the HR PAN image be denoted as $\mathbf{P} \in \mathbb{R}^{H \times W \times 1}$. The MR MS image is represented as $\mathbf{M} \in \mathbb{R}^{h_1 \times w_1 \times c}$, where $H = h_1 * s_1$ and $W = w_1 * s_1$ are the spatial dimension sizes of the MR MS, s_1 is the spatial resolution ratio between the MR HS and HR HS, and$c$ $(c \ll C)$ is the number of MS bands. The observed HS image is denoted as $\mathbf{H} \in \mathbb{R}^{h \times w \times C}$, where $H = h * s$ and $W = w * s$ $(h \ll h_1 \ll H, w \ll w_1 \ll W)$ are the spatial dimension sizes of the HS observation image and s is the spatial resolution ratio between the LR HS and HR PAN. The fused image should have the HR of the HR PAN and the high-fidelity spectral information of the LR HS image. The overview network structure of MGFEI-Net, as shown in Fig. 1, consists of three modules: the HIM, the MGFE module, and the two-stage reconstruction block (TRB). We start by extracting deep features through an unshared triplet feature convolution encoder for the input PAN, MS, and HS, which is achieved by the stacked CBR layers in the convolution structure, where the CBR means the 3×3 standard convolution followed by one batch normalization layer and a ReLU activate function. Thus, three types of feature maps corresponding to three input resolutions are produced, i.e., $\mathbf{f}_p$, $\mathbf{f}_m$, and $\mathbf{f}_h$. Next, these features are fed successively to the HIM and MGFE module for subsequent processing. Details will be given in Sections II-B1 and II-B2.

1) Hybrid-Scale Integrated Module: Two complementary information interaction ways in multisource images have been

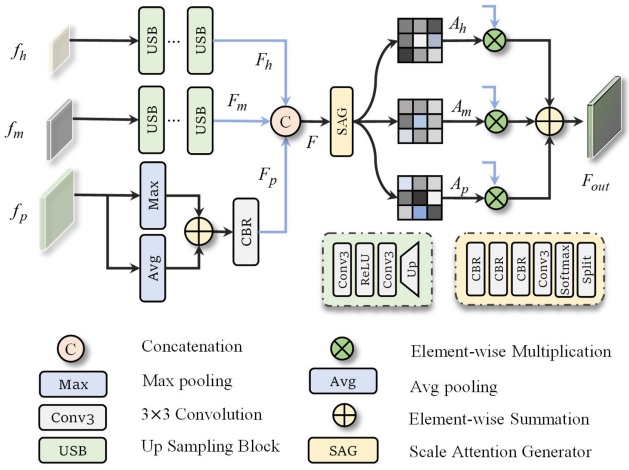


Fig. 2. Illustration of HIM.

widely adopted, i.e., the conventional concatenation of the inputs and then extracting the features or first extracting the multibranch features and then aggregating them. However, since the PAN image shows more geometric and spatial structure cues, the HS image shows more discriminative spectral information, and the MS image has both characteristics, the features of different inputs have different roles in feature aggregation with different adaptive weights. Therefore, the widely used simple interaction strategies may not well calibrate the multisource features to emphasize common scale-invariant objects. Based on the above analysis, we designed an attention-based HIM to screen and combine scale-specific inputs, as shown in Fig. 2. Among the triplet input branches, we adopt a different extraction strategy for each branch. Before feature integration, the input MS and HS image features are first resized to the spatial resolution with the PAN. So, we designed an upsample block (USB), which first fed the image into the convolutional layers to obtain important information, and then upsampling by a factor of 2, as shown in the green background in Fig. 2. For HS, we use a cascading USB to avoid the problem of the spatial and spectral information loss caused by large-scale upsampling (such as 16, 32, and 64) at one time. Specifically, for the PAN branch, we use a hybrid structure of "max pooling and average pooling" to preserve the effective and diverse responses for spatial details in high-resolution features. The processes are

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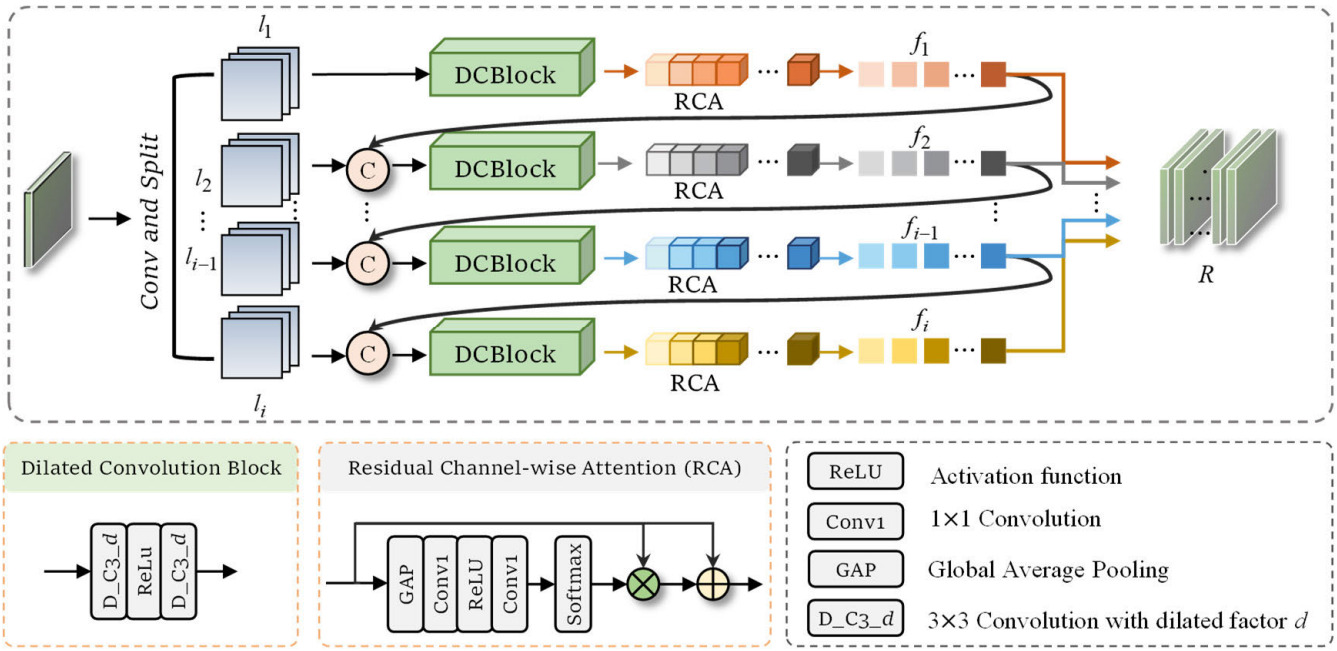


Fig. 3. Schematic illustration of MGFE module.

written as follows:

$$\mathbf{F}_h = \mathrm{USB}_x(\mathbf{f}_h) \tag{1}$$

$$\mathbf{F}_m = \mathrm{USB}_y(\mathbf{f}_m) \tag{2}$$

$$\mathbf{F}_p = \mathrm{CBR}\big(\mathrm{Avg}(\mathbf{f}_p) \oplus \mathrm{Max}(\mathbf{f}_p)\big) \tag{3}$$

where x and y denote the number of stacking USB for MS and HS, respectively.

Then, these features are fed into the HIM to generate attention weights, and we get three feature maps $\mathbf{A}_k$ ($\mathbf{A}_k \in \mathbb{R}^{H\times W\times 1}, k \in \{h, m, p\}$) corresponding to each input feature through a series of convolutional layers. Then, we use the weight map for the final integration to incorporate these characteristics of different input branches into a unified deep structure. The process is formulated as

$$\mathbf{A}_k = \mathrm{softmax}(\mathrm{Conv3}(\mathrm{CBR}(\mathrm{CBR}(\mathrm{CBR}(\mathbf{F})))))) \tag{4}$$

$$\mathbf{F}_{\mathrm{out}} = (\mathbf{A}_h \otimes \mathbf{F}_h) \oplus (\mathbf{A}_m \otimes \mathbf{F}_m) \oplus \big(\mathbf{A}_p \otimes \mathbf{F}_p\big) \tag{5}$$

where softmax denotes the softmax activation function and $\mathbf{F}$ denotes the feature layers that are separately extracted from three branches.

In summary, the HIM can learn the structural uniformity between three inputs based on the attention-aware filtering mechanism, and the learned adaptive coefficient matrix $\mathbf{A}_k$ can be used to mitigate the impact of the multisource imaging mechanism difference to a certain extent and guide the integrated fusion.

2) Multiscale Grouping Feedback Embedded Module: Different from high-level vision problems, HS resolution improvement requires accurate dense pixel prediction. Multiscale features are especially important for remote sensing image interpretation and fusion, and it is difficult to extract multiscale features using 3×3 convolution alone. Therefore, how to promote the network to learn stronger multiscale feature representation and make the network more sensitive to high spatial details without increasing any memory burden are challenging in the fusing process. However, most of the existing works have to increase network depth or introduce the pooling operation or stack more convolution layers to enhance multiscale feature representation ability. As a result, there is a great increase in memory and computational burden. To solve the above problems, inspired by Gao et al. [43], we proposed an MGFE module, which includes the dilated convolution block (DCBlock), RCA block, and the feedback embedded structure, as shown in Fig. 3. Conv and Split indicate that the $\mathbf{F}_{\mathrm{out}}$ as raw input is fed into convolution and equally split into g groups in the channel dimension, denoted by $\mathbf{l}_i (i \in \{1, 2, \ldots, g\})$, each feature subset $\mathbf{l}_i$ has the same spatial size with the input feature map. After the input feature maps $\mathbf{F}_{\mathrm{out}}$ are equally divided into g groups, for every group, we utilize a DCBlock to generate multiscale feature representations and the RCA to enhance important feature channels for producing high discriminative spectral feature. Especially, for every DCBlock consisting of two multiscale dilated operations and one ReLU activation function, as shown in the green background in Fig. 3, utilized the step size of dilated factor d (d is the same as the number of splits grouped layers g) to weight pixels for increasing the receptive field without memory burden and information loss.

The difference in multiscale features mainly lies in that high-level features and low-level features focus on representing semantic context information and spatial detail information, respectively. However, in other grouping multiscale convolutions, all filters of each convolutional layer are individually processed and not fully connected, where there is no between-group communication. In our MGFE, to capture more fine spatial features, the current parts are connected to the input feature of the next group by concatenation operation in

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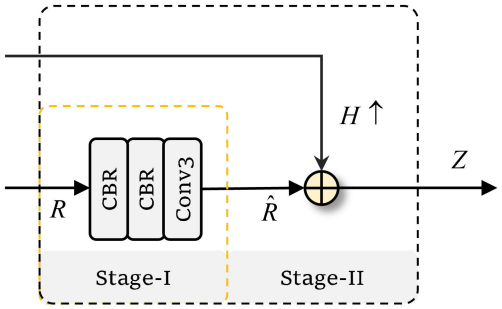


Fig. 4. Illustration of TRB.

the channel dimension for feature reuse, which the high-level features can guide the low-level feature to learn by a top-to-bottom feedback embedded manner with a skip connection, as shown in the long black line in Fig. 3. The whole process can be written as follows:

$$\mathbf{f}_i = \begin{cases} \text{RCA}(\text{DCBlock}(\mathbf{l}_1)), & i = 1 \\ \text{RCA}(\text{DCBlock}(\text{Cat}(\mathbf{f}_{i-1}, \mathbf{l}_i))), & 2 \le i \le g. \end{cases} \tag{6}$$

Finally, the multiple output feature paths of all groups are concatenated in channel dimension and fed into subsequent reconstruct blocks to more sufficiently fuse input features. This process of split and skip connection will be repeated several times until each layer $\mathbf{l}_i$ is processed. $R_\mathbf{i}$ denotes final output feature maps at different feedback embedded stages, which can be written as follows:

$$\mathbf{R_i} = \begin{cases} \mathbf{f}_1, & i = 1 \\ \text{Cat}(\mathbf{R}_{i-1}, \mathbf{f}_i), & 2 \le i \le g. \end{cases} \tag{7}$$

In addition, the parameters of splitting layer number $g$can be controlled to achieve a tradeoff between the effectiveness and efficiency of the multiscale feature extraction, and detailed experimental verification is provided in Section II-B3.

3) Two-Stage Reconstruction Block: After the former MGFE, the fine features $\mathbf{R}$ are fed into the TRB to reconstruct the final HS feature, stage-I has two CBR layers and one 3×3 convolution layer, as shown in the yellow box in Fig. 4. To better reconstruct the spectral information, the original HS image is upsampled and summated with the feature $\hat{\mathbf{R}}$ to improve the spectral information preservation, as shown in stage-II in Fig. 4. After going through the two-stages of the reconstructing block, an ideal HR HS image is obtained. The whole process can be summarized as follows:

$$\hat{\mathbf{R}} = \text{Conv3}(\text{CBR}(\text{CBR}(\mathbf{R}))) \tag{8}$$

$$\mathbf{Z} = \hat{\mathbf{R}} + \mathbf{H} \uparrow \tag{9}$$

where $\mathbf{H} \uparrow$ denotes the upsampling version of the original HS images.

To minimize the penalization between the fusion image and the reference image $\hat{Z}$, the L_1 norm $\mathcal{L}$ is employed as the loss function to better learn a nonlinear mapping relationship

$$\mathcal{L} = \frac{1}{N} \sum_{n=1}^{N} \left\| \hat{Z} - \mathbf{Z} \right\|_1 \tag{10}$$

where N denotes the number of train images.

III. Experiments

A. Datasets

To comprehensively test and verify the performance of the proposed method, the experiments were performed on both simulated and real datasets.

1) Chikusei Dataset [44]: The Chikusei dataset was acquired by the Headwall Hyperspec-VNIR-C imaging sensor in the agricultural and urban areas of Chikusei, Ibaraki Prefecture, Japan. It has 128 bands in the spectral range of 0.363–1.018 μm. The spatial dimension size is 2517 $\times$ 2335, and the spatial resolution is 2.5 m. The main scenes are urban and rural areas. In the experiments, the original data were partitioned into no-overlapped cubic patches of size 256 $\times$ 256 $\times$ 128. The simulated LR HS image was obtained through an 8 $\times$ 8 Gaussian kernel with a standard deviation of 2 and then downsampled by the spatial resolution ratio of 8 [45], [46]. The MR MS image was degraded in both spatial and spectral dimensions. In the spatial dimension, the original HS image was blurred by an 8 $\times$ 8 Gaussian kernel with a standard deviation of 2 and then downsampled by a factor of 2. In the spectral dimension, by dividing the spectral range of the original HS into four parts on average, the MS image of four bands was simulated. The single-band HR PAN image was obtained by averaging the overall spectral range from the original HS image. The dimension sizes of the HR PAN, MR MS, and LR HS images are 256 $\times$ 256, 128 $\times$ 128 $\times$ 4, and 32 $\times$ 32 $\times$ 128, respectively. The training set of the Chikusei dataset contains 50 randomly selected image pairs (about 80%), and retaining 15 image pairs are utilized for testing.

2) Pavia Center Dataset [47]: The image of the Pavia Center dataset, located in Pavia, northern Italy, was obtained by the Reflective Optical System Imaging Spectrometer (ROSIS). The spatial dimension size of the image is 1096 $\times$ 715, and there are 102 bands in the spectral range from 0.430 to 0.860 μm. The spatial resolution is 1.3 m. The major land surface is buildings, roads, and trees.

3) Botswana Dataset [48]: The Botswana dataset was collected on the Okavango Delta in Botswana with a 30-m spatial resolution using the Hyperion sensor mounted on the Earth Observation One (EO-1) satellite. The spatial dimension size is 1496 $\times$ 256, and 242 bands are included in the spectral range from 0.4 to 2.5 μm. In the experiments, 145 bands were retained after removing the bands with noise and moisture absorption characteristics.

For the simulated Pavia and Botswana datasets, the original HS from the above HS datasets was used as the reference images in the simulation experiments. The acquisition process of PAN, MS, and HS images are similar to the Chikusei data, but the size of HS, MS, and PAN is 16 $\times$ 16 $\times$ C(C denotes the number of bands for Pavia and Botswana data), 32 $\times$ 32 $\times$ 4, and 128 $\times$ 128, respectively. For the Pavia dataset, we randomly chose 30 image pairs for training (80%), and the rest of the ten image pairs were used as the testing set. For the Botswana dataset, 22 image pairs for training (80%) were selected at random for training, and the rest of the ten image pairs were used for the testing. In addition,

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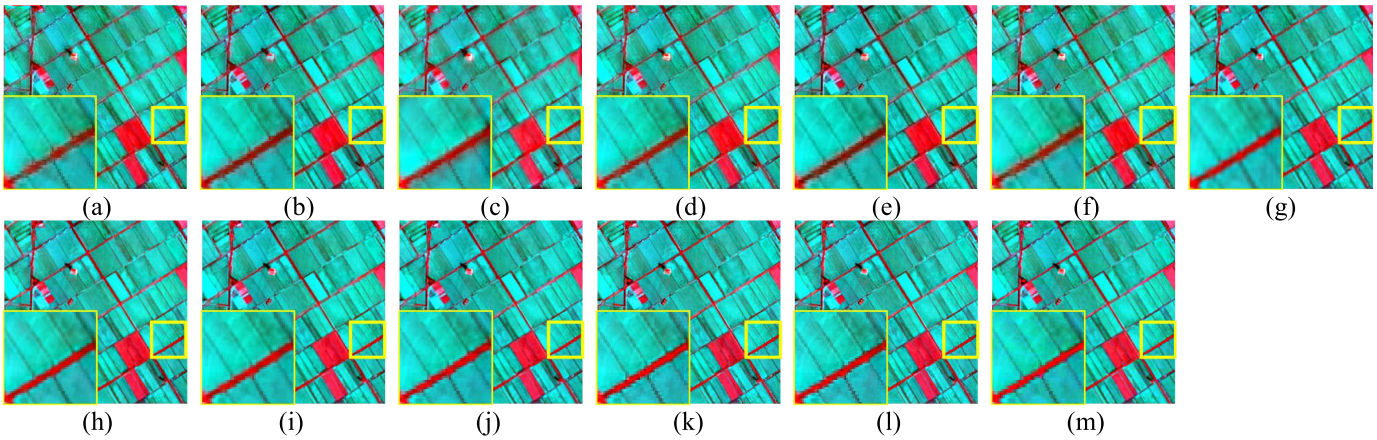


Fig. 5. Visual comparison of fusion results on the simulated Chikusei dataset. (a) FUSE. (b) ICCV15. (c) GLP-HS. (d) GSA. (e) CNMF. (f) ISTSF. (g) PNN. (h) PanNet. (i) HyperPNN. (j) HSpeNet. (k) IFPMH. (l) Proposed. (m) Reference. Large boxes are enlargements of the small boxes.

data augmentation (rotating and flipping) [49] is adopted here to increase the volume of the training set and avoid overfitting.

4) HZB Dataset: In the real experiment, the dataset was collected on the Hangzhou Bay (HZB) area by Zi Yuan (ZY)-1 02D satellite, which provided the LR HS, MR MS, and HR PAN images. The HS image covers the wavelength range of 0.4–2.5 μm with a spatial resolution of 30 m. The original HS (size of 1400 × 900 × 166) consists of 166 spectral bands, and after removing the uncalibrated and noisy bands, the remaining 137 bands were used for our study. The corresponding MR MS (size of 4200 × 2700 × 8) has eight bands and covers the spectral range of 0.452–0.902 μm with a spatial resolution of 10 m. The HR PAN is 0.452–1.047 μm, and its spatial resolution is 2.5 m (size of 16 800 × 10 800). Due to the lack of the ideal reference HR HS, the reduced-resolution training samples were generated according to Wald's protocol [50] for the supervised network. The 80% (about 60 image pairs) region was divided into blocks as the training set, with the remainder used for testing. Specifically, the training size of HS, MS, and PAN is 10 × 10 × 137, 30 × 30 × 8, and 120 × 120, and the testing size of those is 30 × 30 × 137, 90 × 90 × 8, and 360 × 360, respectively. For the full-resolution experimental result, the well-trained model on the reduced-resolution dataset was directly employed for the fusion of the original full-resolution PAN, MS, and HS images.

B. Compared Methods and Evaluation Measures

To verify the performance of the proposed method in the PAN–MS–HS fusion, on the one hand, the proposed method was compared to the existing integrated PAN–MS–HS fusion methods, including the CNMF [31] and ISTSF [51]. The DL-based PNN [16], PanNet [52], HyperPNN [5], and HSpeNet [53] were extended in this article and applied to PAN–MS–HS integrated fusion. Considering the DL-based IFPMH [42] was studied on the integrated fusion of the PAN, MS, and HS images with different swath widths, we removed the zero-in-network module and retrained in our dataset to verify the performance. In addition, we use the FUSE [13], ICCV15 [35], GLP-HS [11], and GSA [54] methods to design a step-by-step strategy to achieve the fusion of the PAN, MS, and HS images, which was divided into two steps. The first step was to implement the MS and HS fusion to get an initial HS fused image with the MS spatial resolution. In the second step, the same method was used to fuse the fusion result of the previous step with the PAN image. The parameters of the above machine learning methods were set mostly following the original studies, including the number of filters and iterations. For the DL-based: PNN and PanNet, the upsampled MS, HS, and original PAN are concatenated on the spectral dimension as input data, which is different from the original setting that only contains PAN and MS images. Besides, for the HyperPNN and HSpeNet methods, we added an MS branch with the same network structure as the PAN branch, and subsequent training was performed in the same manner as before. The training process batch size was set to 4, the number of epochs was 500, and the Adam optimizer [55] with a fixed learning rate of 0.0001 was used to minimize L_1 loss. These parameters were commonly the same for the above DL-based methods, as well as our proposed method [56]. The training and testing of the model were performed by the Pytorch DL framework on the NVIDIA RTX 1080Ti GPU.

To evaluate the performance of our method, seven reduced-resolution quantitative quality indices were employed [4], [25], including universal image quality index (UIQI), spectral angle mapper (SAM), spatial correlation coefficient (SCC), erreur relative globale adimensionnelle de synthèse (ERGAS), peak signal-to-noise ratio (PSNR), structure similarity (SSIM), and correlation coefficient (CC). In addition, to fully demonstrate the advantage of our method in full resolution, we also use the typical nonreference QNR [57] and MQNR [58] as the evaluation criteria on real datasets.

C. Experimental Results Based on Simulated Datasets

The experimental results on the Chikusei dataset, the Pavia Center dataset, and the Botswana dataset are shown in Figs. 5–10, respectively. The quantitative evaluation results are

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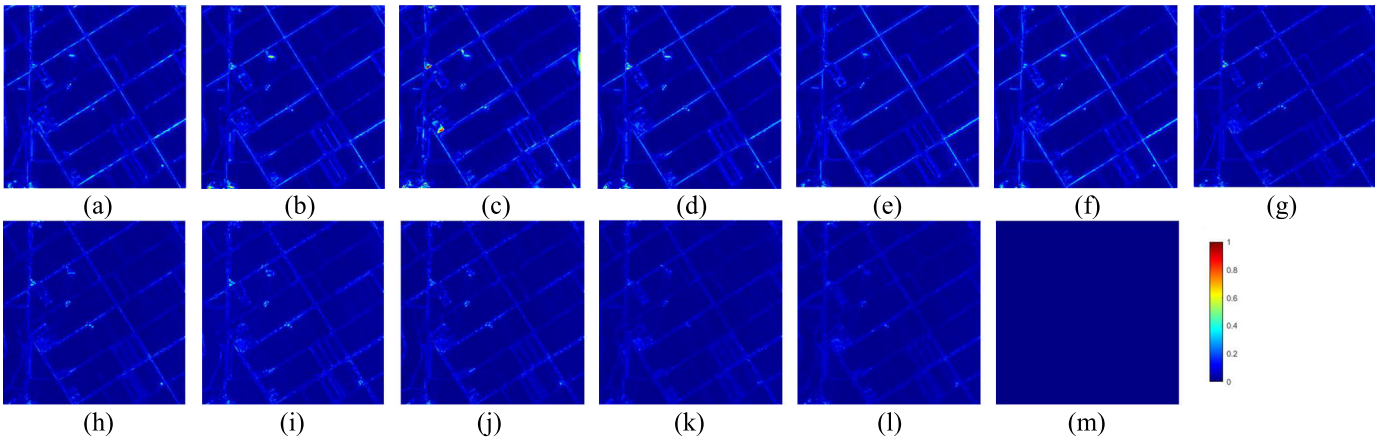


Fig. 6. Spectral error map between the fusion result and the reference image in Chikusei. (a) FUSE. (b) ICCV15. (c) GLP-HS. (d) GSA. (e) CNMF. (f) ISTSF. (g) PNN. (h) PanNet. (i) HyperPNN. (j) HSpeNet. (k) IFPMH. (l) Proposed. (m) Reference.

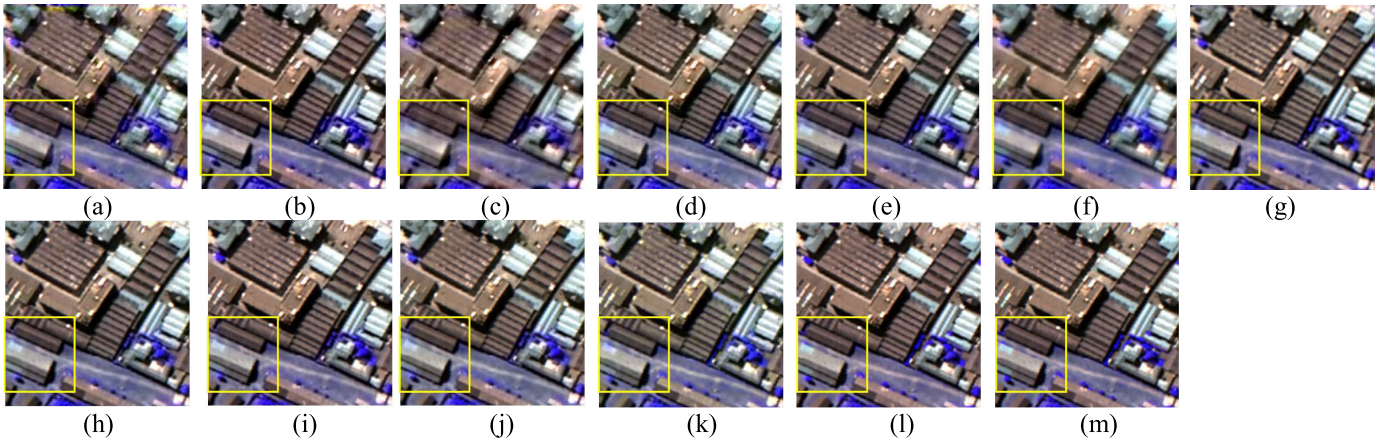


Fig. 7. Visual comparison of fusion results on the simulated Pavia dataset. (a) FUSE. (b) ICCV15. (c) GLP-HS. (d) GSA. (e) CNMF. (f) ISTSF. (g) PNN. (h) PanNet. (i) HyperPNN. (j) HSpeNet. (k) IFPMH. (l) Proposed. (m) Reference.

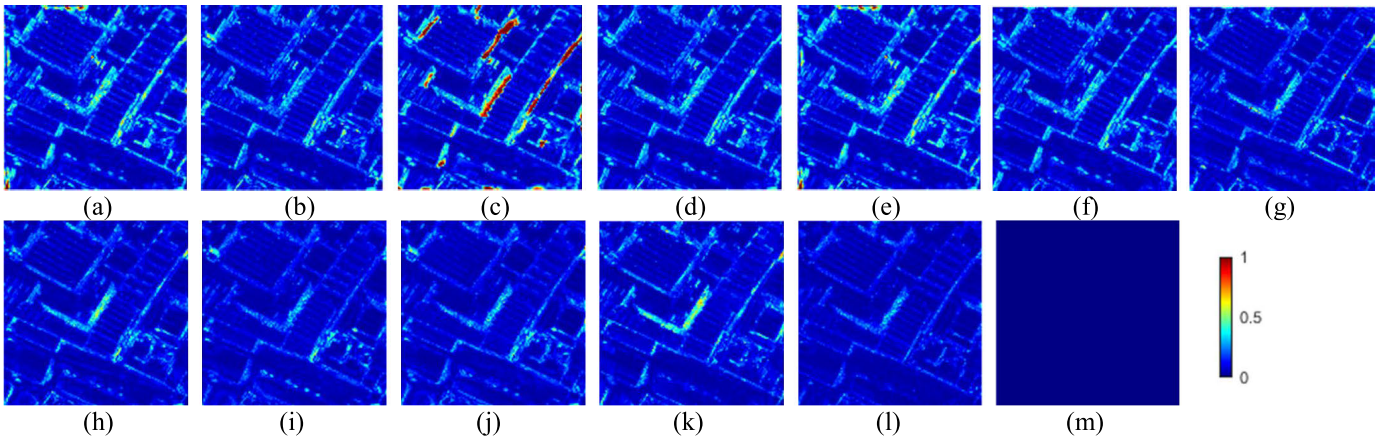


Fig. 8. Spectral error map between the fusion result and the reference image in Pavia. (a) FUSE. (b) ICCV15. (c) GLP-HS. (d) GSA. (e) CNMF. (f) ISTSF. (g) PNN. (h) PanNet. (i) HyperPNN. (j) HSpeNet. (k) IFPMH. (l) Proposed. (m) Reference.

shown in Tables I–III, and the top three results are highlighted in red, green, and blue.

1) Experiments Based on the Chikusei Dataset: Fig. 5 shows the PAN–MS–HS fusion results in the Chikusei dataset. Fig. 5(a)–(k) shows the PAN–MS–HS fusion of the compared methods. The proposed fusion result is shown in Fig. 5(l), and Fig. 5(m) shows the reference image. For the FUSE method, the spectral fidelity of the result is poor, and the phenomenon of transition smoothness occurs for the ICCV15 method. The GLP-HS and GSA methods show relatively poor spectral fidelity and spatial details. Even if the intermediate information MS image is added, the multiplication of the gain factors during fusion will cause serious cumulative errors, and even serious blocking effects, as shown in Fig. 5(c) and (d).

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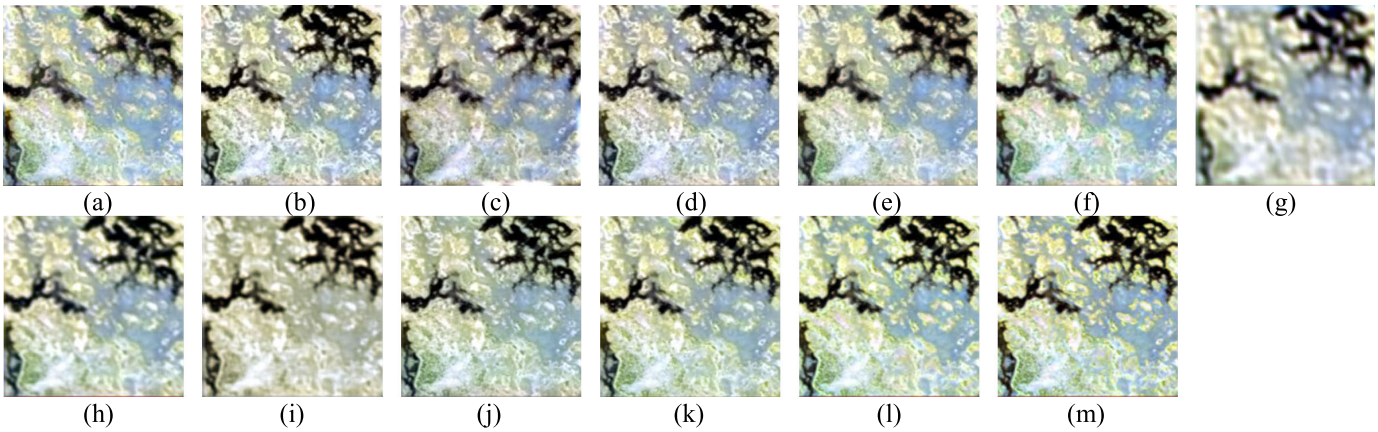


Fig. 9. Visual comparison of fusion results on the simulated Botswana dataset. (a) FUSE. (b) ICCV15. (c) GLP-HS. (d) GSA. (e) CNMF. (f) ISTSF. (g) PNN. (h) PanNet. (i) HyperPNN. (j) HSpeNet. (k) IFPMH. (l) Proposed. (m) Reference.

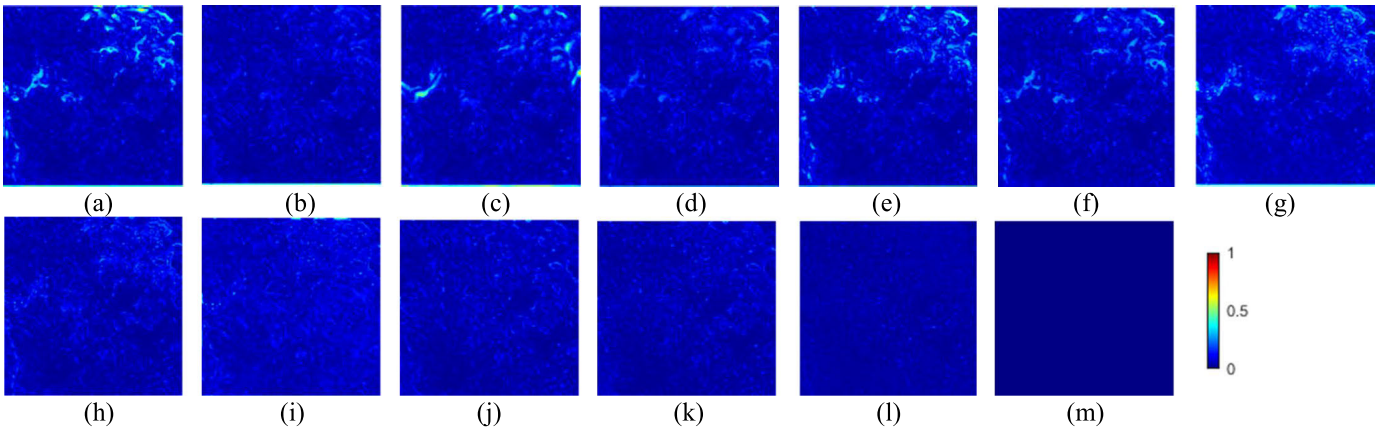


Fig. 10. Spectral error map between the fusion result and the reference image in Botswana. (a) FUSE. (b) ICCV15. (c) GLP-HS. (d) GSA. (e) CNMF. (f) ISTSF. (g) PNN. (h) PanNet. (i) HyperPNN. (j) HSpeNet. (k) IFPMH. (l) Proposed. (m) Reference.

TABLE I

QUANTITATIVE COMPARISON ON THE CHIKUSEI DATASET. ↑ (↓) DENOTES THE LARGER THE BETTER (THE SMALLER THE BETTER). THE TOP THREE RESULTS ARE HIGHLIGHTED IN RED, GREEN, AND BLUE

Method	UIQI ↑	SAM ↓	SCC ↑	ERGAS ↓	PSNR ↑	SSIM ↑	CC ↑
FUSE	0.5882	3.3670	0.6749	3.3864	30.7136	0.8553	0.9221
ICCV15	0.6216	3.3254	0.6910	3.2681	31.9704	0.7895	0.9288
GLP-HS	0.5635	4.1231	0.6405	3.6878	28.3493	0.8466	0.8901
GSA	0.6189	3.3249	0.6557	3.1956	31.2979	0.8724	0.9312
CNMF	0.6463	3.0467	0.7932	2.7064	33.3163	0.9141	0.9465
ISTSF	0.5856	3.2900	0.7024	3.0455	31.2798	0.8606	0.9342
PNN	0.6972	2.8031	0.7149	2.4283	33.5554	0.9127	0.9564
PanNet	0.7533	2.4564	0.7878	2.3151	34.7844	0.9379	0.9639
HyperPNN	0.7457	2.8780	0.8170	2.4510	34.2563	0.9367	0.9580
HSpeNet	0.7713	2.3973	0.8310	2.1518	35.5098	0.9438	0.9676
IFPMH	0.8175	2.0401	0.8464	1.8657	37.1666	0.9548	0.9758
Proposed	0.8321	1.8098	0.8814	1.7045	38.0407	0.9581	0.9791

TABLE II

QUANTITATIVE COMPARISON ON THE PAVIA DATASET. ↑ (↓) DENOTES THE LARGER THE BETTER (THE SMALLER THE BETTER). THE TOP THREE RESULTS ARE HIGHLIGHTED IN RED, GREEN, AND BLUE

Method	UIQI ↑	SAM ↓	SCC ↑	ERGAS ↓	PSNR ↑	SSIM ↑	CC ↑
FUSE	0.7630	7.7257	0.7950	2.7569	29.0309	0.8549	0.9194
ICCV15	0.8258	6.6670	0.7993	2.2190	31.2910	0.7052	0.9468
GLP-HS	0.7087	8.9600	0.7929	3.5823	26.6359	0.8201	0.8704
GSA	0.7943	6.9088	0.8233	2.3632	30.4477	0.8873	0.9465
CNMF	0.8501	6.2879	0.8464	1.9092	32.7230	0.9017	0.9597
ISTSF	0.6961	7.0020	0.7691	2.8192	28.7256	0.8447	0.9249
PNN	0.8520	5.8327	0.7578	2.0068	32.0527	0.9023	0.9596
PanNet	0.8715	5.4788	0.8621	1.8144	33.1939	0.9038	0.9642
HyperPNN	0.8819	5.2630	0.8828	1.6201	34.1933	0.9290	0.9702
HSpeNet	0.8797	5.4106	0.8890	1.6972	33.6679	0.9278	0.9703
IFPMH	0.9073	4.8385	0.9015	1.5203	35.4262	0.9356	0.9883
Proposed	0.9484	3.1247	0.9520	0.8957	41.9783	0.9757	0.9896

The CNMF and five DL-based methods perform well. PNN is only a simple thee-layer network, and the PanNet network adds a ResNet block that is sensitive to high-frequency residuals, PanNet is better than PNN in the fusion result. From the HS-pansharpening method: HyperPNN and HSpeNet, we can see that from Fig. 5(i) and (j), they have better performance in spatial and spectral details than other pansharpening methods. Especially, for the IFPMH, it is a

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TABLE III
QUANTITATIVE COMPARISON ON THE BOTSWANA DATASET. ↑ (↓) DENOTES THE LARGER THE BETTER (THE SMALLER THE BETTER). THE TOP THREE RESULTS ARE HIGHLIGHTED IN RED, GREEN, AND BLUE

Method	UIQI ↑	SAM ↓	SCC ↑	ERGAS ↓	PSNR ↑	SSIM ↑	CC ↑
FUSE	0.7638	2.6132	0.8069	1.1961	26.3116	0.8384	0.9008
ICCV15	0.7908	2.3546	0.8129	1.1413	26.9768	0.8327	0.9086
GLP-HS	0.7749	2.4510	0.8270	1.3802	25.0335	0.8127	0.8692
GSA	0.8080	2.3302	0.8433	1.0454	27.4886	0.8803	0.9287
CNMF	0.7542	2.6622	0.7953	1.2406	26.4364	0.8654	0.9044
ISTSF	0.6157	2.4333	0.6547	1.4377	25.4579	0.7749	0.8687
PNN	0.4383	3.1481	0.6104	1.9922	22.5798	0.5515	0.8206
PanNet	0.7627	2.1664	0.8549	0.9125	28.8196	0.8226	0.9401
HyperPNN	0.6829	3.1231	0.6908	1.4676	24.7830	0.7655	0.8578
HSpeNet	0.8167	2.2593	0.8520	1.1763	27.6836	0.8674	0.9222
IFPMH	0.8616	1.6431	0.9332	0.8243	29.2802	0.8918	0.9541
Proposed	0.8708	1.6429	0.9362	0.7683	31.9479	0.9012	0.9671

specially designed integrated fusion of the PAN, MS, and HS images, which have more competitive performance than other methods, as shown in Fig. 5(k). It can be seen from Fig. 5(l) that the proposed method has good performance in both spatial detail information enhancement and spectral information preservation.

To further verify the performance of the proposed method on the Chikusei dataset, we show the spectral angle error map of each pixel between the fused image and the reference image, as shown in Fig. 6. The spectral distortion gradually increases from 0 to 1, and the color changes from blue to red. The smaller the value, the better the fusion result. From Fig. 6, the proposed method can generate the smallest error of the fusion result in dense buildings and other areas with rich spatial texture information, which proves the good performance of our proposed method.

The statistical average quantitative results are shown in Table I. It can be seen that the proposed method is better than all other methods on UIQI, SAM, CC, ERGAS, PSNR, SSIM, and CC, indicating that the proposed method has the best fusion performance among all competing methods.

2) Experiments Based on the Pavia Center Dataset: Fig. 7 shows different fused results on the Pavia Center dataset. Among them, Fig. 7(a)–(k) shows the fusion of the other compared methods, the proposed method is shown in Fig. 7(l). For the GSA and GLP-HS methods, the GSA is better than GLP-HS in terms of spatial details and spectral preservation performance. The ISTSF and FUSE methods are poor in spatial details, and also, both show the phenomenon of spectral distortion, especially for the FUSE method. The CNMF method is shown in Fig. 7(e), which performs well in terms of spectral preservation and spatial information among all comparison methods except those based on DL. Although PNN, PanNet, HyperPNN, HSpeNet, IFPMH, and our method have roughly the same ability to visualize the spatial details of the appearance of the map, they are not as good as our proposed method in terms of preserving spectral information in quantitative indicators. To further prove the effectiveness of the proposed method, the spectral angle error images are shown in Fig. 8, where blue represents a small difference and red represents a serious spectral distortion. The proposed method has the smallest error value among all compared methods, indicating that the fusion result is closest to the reference image. Table II lists the statistical average quantitative results in the Pavia Center dataset. The fusion accuracy of our proposed method is the best among all indicators, which indicates that the proposed method can fulfill the PAN–MS–HS fusion with the least distortion on spatial detail and spectral fidelity.

3) Experiments Based on Botswana Dataset: For the Botswana dataset, we give the statistical average quantitative results of all testing images in Table III. It can be seen from Table III that our proposed method has once again obtained the best value in all evaluation indicators, surpassing the traditional method and the HS sharpening method based on the CNN. Fig. 9 shows the fusion performance of different methods in the Botswana dataset. It can be seen from Fig. 9(a) to (l) that the FUSE, ICCV15, GLP-HS, and GSA methods all have spatial and spectral distortions. It was unexpected that the DL-based PNN and HyperPNN methods do not perform as well as traditional methods. Compared with the CNMF method in the Chikusei dataset and the Pavia Center dataset fusion performance, it produced unexpected results. This can be proved by the evaluation index table and the visualization image. Like the previous experiments, we show the spectral angle error between the fused image and the reference image, as shown in Fig. 10. In general, corresponding to the visualization results, it is obvious that the proposed method obtains the fusion image with minimal error, which shows that our proposed method has high spectral fidelity.

D. Experimental Results Based on Real Datasets

To further demonstrate the superiority of the proposed methods, the real ZY-1 02D PAN–MS–HS remote sensing dataset collected in the south of HZB on January 29, 2021, was used. The fusion results are shown in Fig. 11. To evaluate the spectral preservation capability of different fusion methods, Fig. 12 shows the spectral performance. Among them, Fig. 12(a)–(c) shows the spectral value of different fusion results and the reference image on coordinates (200, 200), (120, 120), and (300, 90) of the representative surface features. As depicted in Fig. 12, a black line in Fig. 12(a)–(c) served as the benchmark, which means that the closer to the benchmark line, the less spectral distortion of the fusion results. From Fig. 12, our proposed method obtains the best approximation to the reference HS in the spectral domain, which demonstrates the superior spectral fidelity preservation of our proposal. To prove the performance of the proposed method, we apply the trained model to a full-resolution dataset, visualize the results, and the statistical quantitative evaluations are listed in Table IV. Combined with the numerical evaluations and visual effects, it can be still concluded that our proposed

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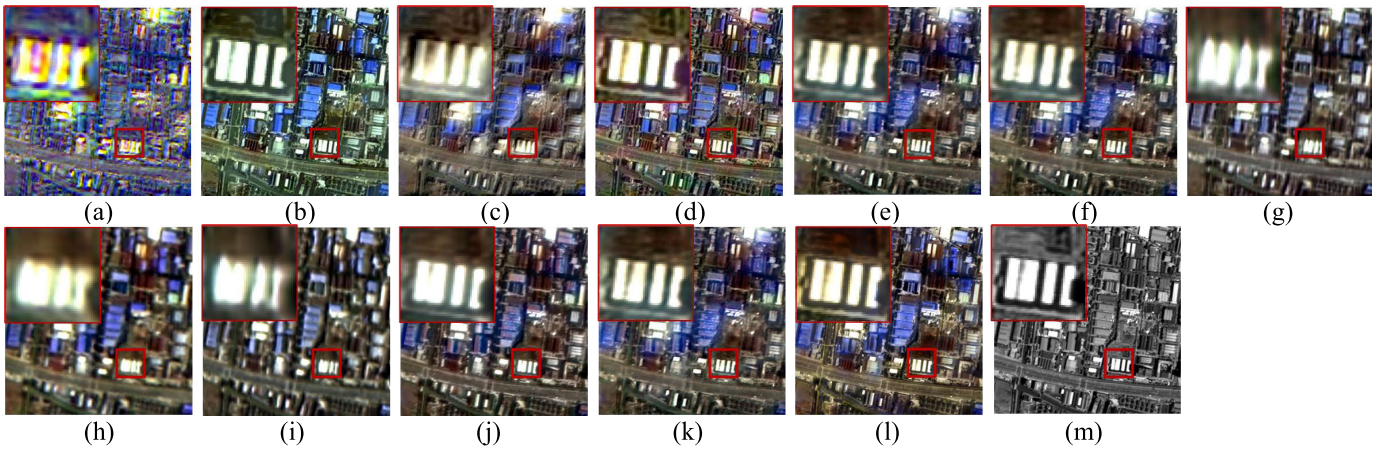


Fig. 11. Visual comparison of fusion results on the real HZB dataset. (a) FUSE. (b) ICCV15. (c) GLP-HS. (d) GSA. (e) CNMF. (f) ISTSF. (g) PNN. (h) PanNet. (i) HyperPNN. (j) HSpeNet. (k) IFPMH. (l) Proposed. (m) PAN.

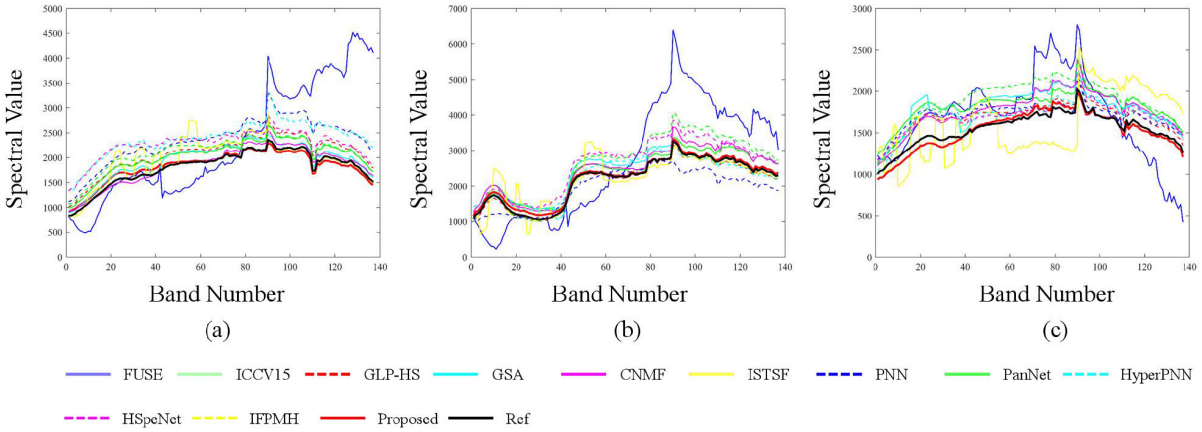


Fig. 12. Comparison of spectral values of three representative locations in the HZB dataset on the single pixel. (a) Pixel (200, 200). (b) Pixel (120, 120). (c) Pixel (300, 90).

TABLE IV

QUANTITATIVE COMPARISON ON HZB DATASETS. ↑ (↓) DENOTES THE LARGER THE BETTER (THE SMALLER THE BETTER). THE TOP THREE RESULTS ARE HIGHLIGHTED IN RED, GREEN, AND BLUE

Method	FUSE	ICCV15	GLP-HS	GSA	CNMF	ISTSF	PNN	PanNet	HyperPNN	HSpeNet	IFPMH	Proposed
QNR ↑	0.7117	0.8253	0.8399	0.8365	0.8273	0.8263	0.8596	0.8451	0.8379	0.8523	0.8536	0.8645
MQNR ↑	52.9314	75.2804	79.5600	87.1189	172.3836	102.3836	115.3832	139.9503	183.3145	204.7073	214.2527	250.5941

methods have the best performance among all comparison methods.

E. Ablation Study

To evaluate the contribution of different modules in the proposed MGFEI-Net, the ablation experiments were performed on the Chikusei dataset, as shown in Table V. We utilized the proposed method as a benchmark for comparison. There are two critical ingredients in the proposed model: the HIM, and the MGFE. In MGFE, the grouping number G, the RCA, the skip connection, and the DCBlocks were also conducted for verifying the performance.

1) Effectiveness of the HIM Module: To demonstrate the effectiveness of the HIM, we replace the HIM with a simple concatenation for PAN, MS, and HS in the channel dimension.

TABLE V

AVERAGE QUANTITATIVE RESULTS OF DIFFERENT BLOCK DESIGNS ON THE CHIKUSEI DATASET

Method	UIQI ↑	SAM ↓	SCC ↑	ERGAS ↓	PSNR ↑	SSIM ↑	CC ↑
w/o HIM	0.7724	2.3257	0.8361	2.1264	35.0679	0.9475	0.9675
w/o MGFE	0.6294	3.2411	0.6562	3.6854	30.5730	0.8450	0.9174
w/o RCA	0.8317	2.1390	0.8801	1.7280	37.9672	0.9445	0.9768
w/o S_C	0.8299	1.9378	0.8748	1.7153	38.0199	0.9564	0.9788
w/o DCBlock	0.8155	2.0530	0.8443	1.8840	36.7589	0.9539	0.9755
Proposed	0.8321	1.8098	0.8814	1.7045	38.0407	0.9581	0.9791

It can be seen from Table V, we assert that HIM can effectively explore and fuse complementary information between the

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TABLE VI
AVERAGE QUANTITATIVE RESULTS OF DIFFERENT GROUPING NUMBERS IN THE MGFE

Grouping Number G	UIQI ↑	SAM ↓	SCC ↑	ERGAS ↓	PSNR ↑	SSIM ↑	CC ↑
2	0.8126	2.0747	0.8402	1.9300	36.9510	0.9526	0.9744
3	0.8169	2.0255	0.8571	1.8610	37.3283	0.9549	0.9759
4	0.8321	1.8098	0.8884	1.7045	38.0407	0.9581	0.9791
5	0.8297	1.8855	0.8843	1.7987	38.0016	0.9559	0.9781
6	0.8074	1.9863	0.8690	1.8358	37.9534	0.9534	0.9760

spatial and spectral features among the PAN, MS, and HS images.

2) Effectiveness of the MGFE Module: To confirm that the MGFE module can effectively bridge the intrinsic multiscale and enhance the contact between cross-level features, we introduced the ablation experiments for MGFE as follows.

a) The other ingredients of the MGFE: To explore whether ingredients can improve the performance of MGFE, we deleted the DCBlock, the RCA, and the Skip connection, respectively. As shown in Table V, if any structure is deleted, the performance of the module is reduced. Among them, the DCBlock module has a greater impact on performance. In addition, we also replaced the MGFE with a simple stack convolution layer as shown in Table V. It shows that the performance of the experiment is degraded largely, which indicates that the proposed MGFE module can effectively improve the reconstruction performance and the fusion accuracy.

b) Grouping number: To investigate the effect of the number of groups, we report the variation of quantitative results and training time cost versus, only the numbers of grouping changing on the Chikusei dataset. As shown in Table VI, the average quantitative results, i.e., UIQI, SCC, SAM, ERGAS, PSNR, SSIM, and CC can be achieved best when $G = 4$. Especially, when the number of grouping numbers increases from 5 to 6, the training time of the network grows rapidly, but the quantitative results are dropped. Finally, considering both the fusion performance and training time cost, we choose the number of groupings to be set to 4, which can obtain the best fusion performance while reducing the time cost.

F. Network Complexity

In DL models, in addition to comparing the subjective visual effects and the objective indicators, complexity is also an important indicator. Therefore, the commonly used total number of parameters, the number of floating-point operations (FLOPs) trained in the network, and the network test time were selected to evaluate the complexity of the DL models. Table VII shows the results of the complexity indicators on the Chikusei dataset. It shows that the parameters and FLOPs of our proposed network are larger than other DL-based methods, and it has similar performance to the DL-based methods in the test time. In the future, the network complexity and computational efficiency should be further considered.

IV. DISCUSSION

A. PAN–MS–HS Fusion Versus PAN–HS Fusion

To verify the advantage of the proposed PAN–MS–HS fusion, two versions of the proposed fusion method with all the PAN, MS, and HS images and the direct fusion with only PAN and HS images were performed. The proposed method was compared with existing state-of-the-art PAN–HS fusion methods, including the FUSE [13], the ICCV15 [35], the GLP-HS [11], the GSA [54], the CNMF [23], and four DL-based methods, i.e., the PNN [16], PanNet [52], HyperPNN [5], and HSpeNet [53]. The spatial resolution ratio between the PAN and HS was 8 times. The experimental results based on the Chikusei dataset are shown in Fig. 13. Among them, Fig. 13(a)–(i) represents the fusion results of the compared methods. Fig. 13(j) is the proposed fusion result with PAN and HS images, and Fig. 13(k) is the proposed fusion result with the PAN, MS, and HS images. Fig. 13(l) denotes the reference image. The quantitative evaluation results are shown in Table VIII.

In Fig. 13, the local areas are magnified with red squares for convenient comparison. It is observed that the FUSE and ICCV15 methods have some loss of spatial details, especially in the edge and texture areas. Fig. 13(e) shows that the CNMF method has spectral distortion in some areas. The GLP-HS and GSA have better performance in texture detail. The four DL-based methods, i.e., PNN, PanNet, HyperPNN, and HSpeNet, achieved better spatial detail reconstruction and spectral information preservation than the traditional method.

Compared with the reference image, the proposed fusion method with only PAN and HS images and all the PAN, MS, and HS images shows the best results than the other method. However, the PAN–MS–HS fusion result is more similar to the label image than the PAN–HS fusion, which can also be seen in Fig. 13(j) and (k). The quantitative evaluation in Table VIII shows that, on the one hand, the proposed version with only PAN and HS image fusion achieves a competitive performance than other state-of-the-art PAN and HS fusion methods, which indicates the superior performance of the proposed model. On the other hand, the proposed version with all PAN, MS, and HS obtains the highest indicators on UIQI, SCC, PSNR, SSIM, and CC, and the smallest values of ERGAS, and SAM. This indicates that the proposed method can make full use of the complementary advantages of the HR PAN, MR MS, and LR HS images, and it has the best fusion performance on spatial enhancement and spectral information preservation among all the methods.

B. Analysis of Robustness to Different Spatial Resolution Ratios

To evaluate the spectral fidelity and robustness of different methods to different large spatial resolution ratios, the statistical average quantitative evaluation indices of the PAN–MS–HS fusion results on the Chikusei dataset under different spatial resolution ratios with the 8, 16, 32, and 64 are shown in Fig. 14. Among them, each figure represents the trend of the fusion results at an evaluation index for different methods

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TABLE VII

COMPARISON OF THE NUMBER OF PARAMETERS, FLOPS, AND TEST TIME OF THE COMPARED NETWORK. THE BEST RESULTS ARE HIGHLIGHTED IN RED

Method	FUSE	ICCV15	GLP-HS	GSA	CNMF	ISTSF	PNN	PanNet	HyperPNN	HSpeNet	IFPMH	Proposed
Parameters (MB)	--	--	--	--	--	--	0.97	1.06	0.13	0.19	184.65	5.96
FLOPs (GB)	--	--	--	--	--	--	57.93	68.45	9.08	12.87	200.12	296.70
Test times (s)	3.5822	61.1948	19.7359	12.4869	11.6090	2322.6645	0.5663	0.5525	0.5939	0.3271	10.3738	0.5884

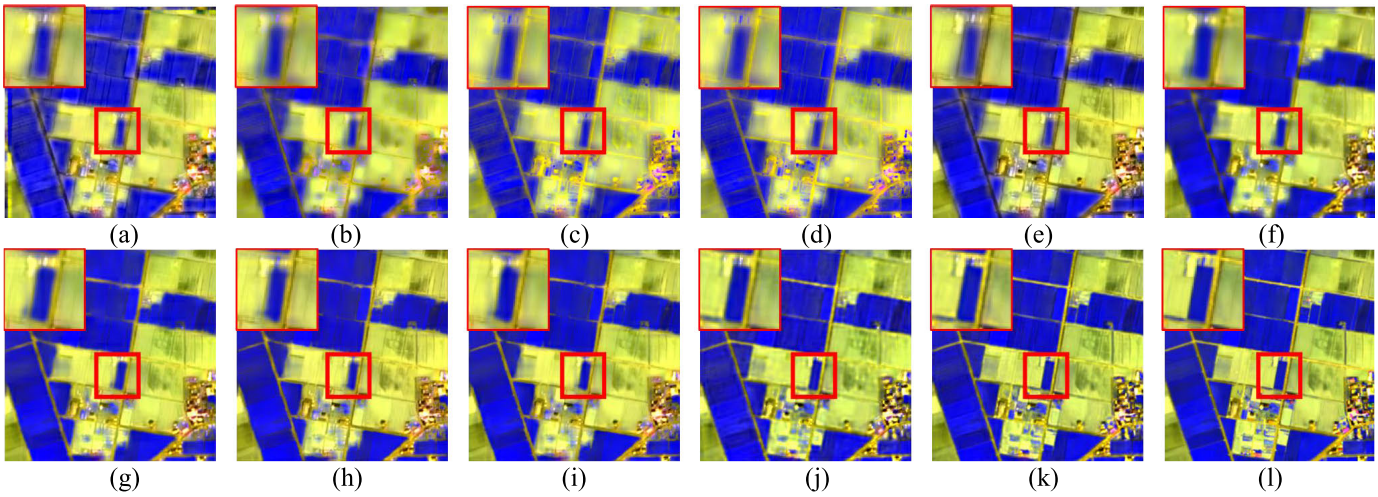


Fig. 13. Analysis of the PAN–HS fusion versus PAN–MS–HS fusion on the Chikusei dataset. (a)–(i) Different compared fusion methods with PAN–HS images. (a) FUSE. (b) ICCV15. (c) GLP-HS. (d) GSA. (e) CNMF. (f) PNN. (g) PanNet. (h) HyperPNN. (i) HSpeNet. (j) Proposed method with PAN–HS images. (k) Proposed method with PAN–MS–HS images. (l) Reference.

TABLE VIII

QUANTITATIVE ANALYSIS OF THE PAN–MS–HS FUSION VERSUS PAN–HS FUSION ON THE CHIKUSEI DATASET. ↑ (↓) DENOTES THE LARGER THE BETTER (THE SMALLER THE BETTER). THE TOP THREE RESULTS ARE HIGHLIGHTED IN RED, GREEN, AND BLUE

Method	UIQI ↑	SAM ↓	SCC ↑	ERGAS ↓	PSNR ↑	SSIM ↑	CC ↑
FUSE (PAN-HS)	0.45197	4.0553	0.6890	3.8104	30.3551	0.7957	0.9020
ICCV15 (PAN-HS)	0.5578	3.9030	0.6567	3.9209	30.4458	0.7798	0.8965
GLP-HS (PAN-HS)	0.5106	4.5479	0.6258	4.1772	28.2561	0.8168	0.8722
GSA (PAN-HS)	0.5427	4.1423	0.6333	4.0241	30.0304	0.8227	0.8926
CNMF (PAN-HS)	0.5556	3.7588	0.7763	3.4408	31.2459	0.8832	0.9146
PNN (PAN-HS)	0.5788	3.5895	0.6710	3.3191	30.9412	0.8851	0.9221
PanNet (PAN-HS)	0.6741	3.0918	0.7794	2.9379	32.3182	0.9188	0.9407
HyperPNN (PAN-HS)	0.6716	3.4206	0.7896	3.1688	31.9892	0.9206	0.9318
HSpeNet (PAN-HS)	0.6805	3.1119	0.8032	2.9724	32.5691	0.9234	0.9400
Proposed (PAN-HS)	0.7653	2.2747	0.8317	2.1048	36.0551	0.9463	0.9695
Proposed (PAN-MS-HS)	0.8321	1.8098	0.8814	1.7045	38.0407	0.9581	0.9791

under different resolution ratios, which denotes the stability of the method. It is shown that the proposed method, namely MGFEI-Net, has overall the best performance compared to other methods in terms of the seven quantitative evaluation indices, i.e., the UIQI, SCC, SSIM, PSNR, ERGAS, SAM, and CC. For the fusion performance under different spatial resolution ratios of 8, 16, 32, and 64, the performance decrease is obvious for the traditional methods, which indicates worse stability to large spatial resolution ratios. For comparison, our proposed method not only leads all other methods in terms of indicators but also performs more stability as the resolution difference increases.

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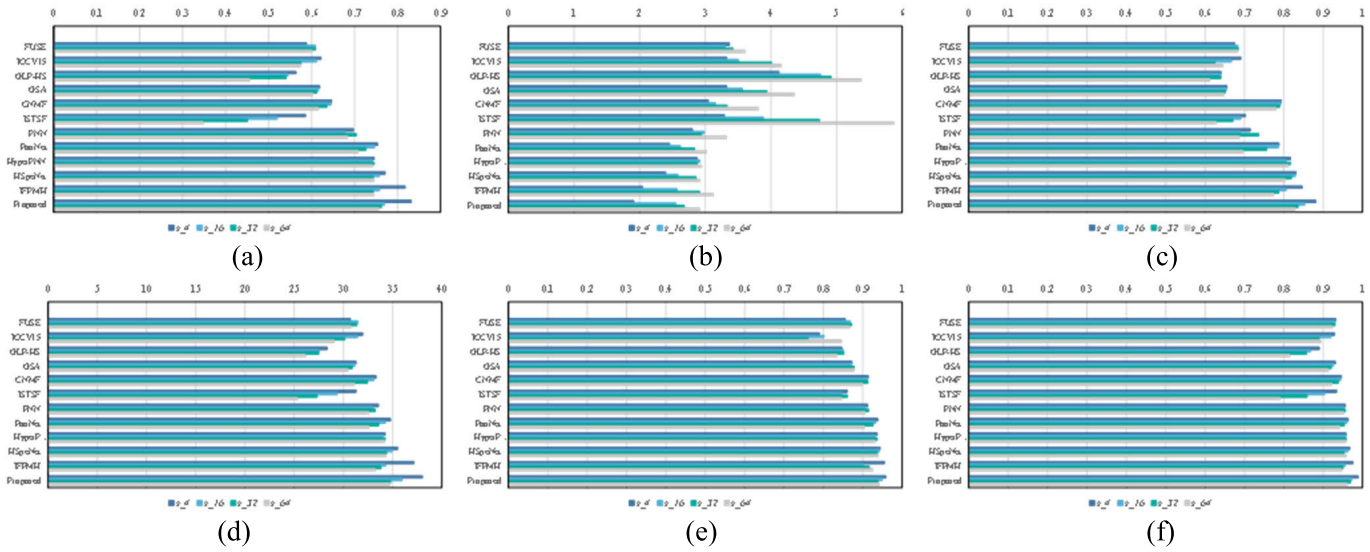


Fig. 14. Statistical average quantitative evaluation indices of the PAN–MS–HS fusion results on the Chikusei dataset under different spatial resolution ratios with 8, 16, 32, and 64, where s denotes the resolution ratio. (a) UIQI value. (b) SAM value. (c) SCC value. (d) PSNR value. (e) SSIM value. (f) CC value.

V. Conclusion

In this article, we propose a multiscale grouping feedback embedded integrated network for the LR HS, MR MS, and HR PAN images by fully exploring their complementary spatial and spectral information. In the proposed method, an attention-based HIM and an MGFE module with a top-to-bottom manner are designed. The experiments were implemented on the simulated and real datasets, including the opening Chikusei dataset, Pavia Center dataset, and Botswana dataset in the simulated experiments, and the real spaceborne ZY-1 02D PAN, MS, and HS images. The experimental results show that the proposed method can effectively integrate the complementary spatial and spectral information of the HR PAN, MR MS, and LR HS images, to obtain the high-fidelity HR HS image. Moreover, the proposed method is robust to different large spatial resolution ratios, and it has the potential to significantly improve the spatial resolution of the LR HS image.

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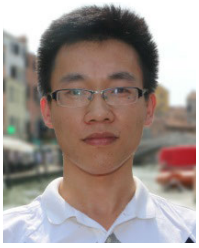
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